

- `$center` (defines a circle)

Points within a Circle Defined on a Sphere

MongoDB supports rudimentary spherical queries on flat 2d indexes for legacy reasons. In general, spherical calculations should use a `2dsphere` index, as described in [2dsphere Indexes](#) (page 496).

To query for legacy coordinate pairs in a “spherical cap” on a sphere, use `$geoWithin` with the `$centerSphere` operator. Specify an array that contains:

- The grid coordinates of the circle’s center point
- The circle’s radius measured in radians. To calculate radians, see [Calculate Distance Using Spherical Geometry](#) (page 537).

Use the following syntax:

```
db.<collection>.find( { <location field> :
  { $geoWithin :
    { $centerSphere : [ [ <x>, <y> ] , <radius> ] }
  } } )
```

The following example query returns all documents within a 10-mile radius of longitude 88 W and latitude 30 N. The example converts distance to radians by dividing distance by the approximate equatorial radius of the earth, 3963.2 miles:

```
db.<collection>.find( { loc : { $geoWithin :
  { $centerSphere :
    [ [ 88 , 30 ] , 10 / 3963.2 ]
  } } } )
```

Proximity to a Point on a Flat Surface

Proximity queries return the 100 legacy coordinate pairs closest to the defined point and sort the results by distance. Use either the `$near` operator or `geoNear` command. Both require a 2d index.

The `$near` operator uses the following syntax:

```
db.<collection>.find( { <location field> :
  { $near : [ <x> , <y> ]
  } } )
```

For examples, see `$near`.

The `geoNear` command uses the following syntax:

```
db.runCommand( { geoNear: <collection>, near: [ <x> , <y> ] } )
```

The `geoNear` command offers more options and returns more information than does the `$near` operator. To run the command, see `geoNear`.

Exact Matches on a Flat Surface

Changed in version 2.6: Previously, 2d indexes would support exact-match queries for coordinate pairs.

You cannot use a 2d index to return an exact match for a coordinate pair. Use a scalar, ascending or descending, index on a field that stores coordinates to return exact matches.